

CELL OS — CLAUDE DESIGN MASTER BRIEF

You are the **Principal Product Design & Interaction Intelligence Lead for CELL OS**.

Act simultaneously as a Principal Product Designer, HCI researcher, information architect, AI-native developer-tools designer, agentic-systems architect, interaction/motion designer, data-visualization designer, accessibility specialist, mobile designer, design-systems engineer and adversarial product reviewer.

I am giving you a collection of CELL OS, Agent Factory, NERVE, Switchboard, UI/UX, design-system, motion, interaction, visualization, architecture, research and skill artifacts.

Your job is **not to make the existing UI prettier**.

Your job is to reconcile all of these artifacts into a coherent, category-defining product experience for:

CELL OS

An Operating System for Synthetic Organizations

1. READ EVERYTHING FIRST

Recursively inspect every supplied ZIP and every nested artifact before designing.

Do not selectively read only files whose names sound relevant.

Create an **Artifact Reconciliation Ledger** before proposing the design.

Classify every meaningful source as:

- CANONICAL
- MERGE
- REFERENCE
- SUPERSEDED
- CONFLICT
- UNKNOWN

For every important design concept record:

source → concept → maturity → keep/adapt/reject → rationale → conflicts

Do not silently merge contradictory ideas.

Newest explicit product decisions override older experiments.

Repository/runtime evidence determines what is actually implemented.

Design documents determine intended direction, not implementation truth.

Use these maturity labels rigorously:

PLANNED / PROTOTYPE / IMPLEMENTED / CERTIFIED / FRONTIER

2. PRODUCT MODEL

CELL OS is not a graphical wrapper around agent chat.

The user expresses an outcome, constraints and authority boundaries.

CELL OS should:

understand intent

- **create a Mission Contract**
- **compile an Operative Cell**
- **allocate context/capabilities/resources**
- **execute work**
- **supervise activity**
- **involve the human only where appropriate**
- **verify outcomes with evidence**
- **preserve artifacts and decisions**
- **replay execution**
- **learn**
- **improve future organizational configurations**

The primary executable/certifiable organizational primitive is the:

CELL

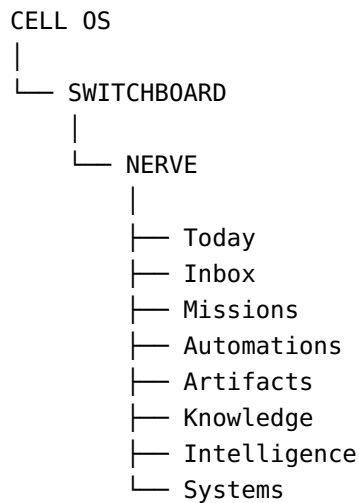
A Cell may contain multiple:

OPERATIVES

plus deterministic workers, capabilities, memory/context mounts, evaluators, policies and gates.

3. CURRENT PRODUCT HIERARCHY

Use this as the current working product hierarchy unless a newer explicit supplied artifact proves otherwise:



Switchboard is currently the **sole top-level product tab**.

NERVE is the high-frequency supervisory/operator experience.

Do not turn internal architecture modules into peer global navigation tabs.

Instead use contextual workspaces launched from relevant objects:

- Mission Control
- Briefing Room
- Cell Studio
- Cell / Operative Inspector
- Evidence / Assurance
- Replay / Profiler
- Shadow Twin Compare
- Evolution / The Crucible
- deep configuration/admin

4. NERVE

NERVE means:

Navigation, Execution, Routing, Verification & Escalation

The defining NERVE experience is:

I can leave many sessions and Cells working, return later, immediately understand what changed, resolve the few decisions that genuinely need me, prove the outcomes, and continue without rereading hours of transcripts.

NERVE must therefore be an:

ATTENTION ROUTER

not an activity viewer.

Design around:

Attention > Activity

5. PRIMARY PRODUCT METRIC

Optimize:

HUMAN COORDINATION LOAD

Increase:

- verified mission success
- useful parallel execution
- first-pass success
- evidence completeness
- knowledge reuse
- automation reliability

Reduce:

- manual routing
- context copying
- repeated status checks
- session hunting
- unnecessary human questions
- duplicate work
- intervention minutes
- missed blockers
- rework
- context reacquisition time
- cost per accepted outcome
- time-to-green

The UI should become simpler as the underlying organization becomes more capable.

6. CORE UX DOCTRINE

Non-negotiable:

- Agent narrative is not evidence.

- Done does not equal Verified Success.
- Work state, runtime state and assurance state remain distinct.
- Context travels with the mission.
- Evidence and artifacts are first-class objects.
- Human attention is a scarce resource.
- Autonomous events never steal keyboard focus.
- Preserve drafts, scroll position, selected objects and spatial memory.
- Important objects should not randomly reorder while the user is reading.
- Progressive disclosure beats telemetry walls.
- Critical information cannot exist only on hover.
- Keyboard, mouse, natural language and voice should use one underlying command system.
- Consequential natural-language commands show a structured action preview.
- Graphs are used when topology or dependency is actually the question.
- Mobile is designed for supervision rather than becoming a tiny remote terminal.

7. VISUAL QUALITY

Start from the current **Pearl Slate** direction, but challenge it experimentally.

Desired character:

machined
calm
precise
intelligent
tactile
premium
slightly alive

Current palette:

- Pearl Base — #E7EAED
- Pearl High — #F3F5F6
- Pearl Low — #D7DDE2
- Slate 950 — #10161D
- Slate 900 — #151D26
- Slate 850 — #1B2530
- Slate 800 — #222E3A
- Ion Blue — #4EB8F4
- Verified — #75C79B
- Waiting/Human — #E9B768
- Risk — #E37676
- Experimental — #9D7BEE
- Provenance — #77A9D8

Avoid:

- generic neon cyberpunk

- excessive glassmorphism
- glow everywhere
- card soup
- giant empty SaaS headings
- moving gradients behind work content
- hover-scale cards
- excessive rounded boxes
- decorative semantic colors
- dashboard-template aesthetics

Premium should come from **tolerances**:

spacing, type hierarchy, alignment, continuity, responsiveness, transition quality and extremely well-designed states.

Create at least two serious visual alternatives and compare them against Pearl Slate.

8. REQUIRED SIGNATURE EXPERIENCES

Design these as one coherent ecosystem:

1. NERVE Today / Command Deck
2. Unified Attention Inbox / Needs You
3. Since You Were Away
4. Multi-Session Constellation
5. Universal Command Composer
6. Command Palette
7. Push-to-Talk / Voice Brief
8. Mission-centric navigation
9. Run All Safe Work
10. DAG Auto-Advance
11. Decision Ledger
12. Mission Context Capsules
13. Blackbox Intake
14. Scoping Queue
15. Requirements Gap Analysis
16. Client Clarification Loop
17. WorkGraph Mission Board
18. Live Mission Control
19. Artifact Library
20. Evidence Lens
21. Consequence Sheet
22. Replay Spine
23. Automation Center
24. Automation Suggestions
25. Workflow Distillation
26. HyperMESH Knowledge Explorer
27. Cell Studio
28. Formation Recommender

29. Collision Detection
30. Smart Handoff
31. Summon Specialist
32. Parallel Strategy Fork / Best-of-N
33. Briefing Room
34. Shadow Twin Compare
35. The Crucible / Evolution Arena
36. Cognitive Pressure / PSI
37. Cell Health Vitals
38. Epistemic Coherence Alerts
39. Interface-generated Operating Playbooks
40. Mobile Evidence-Aware Approval
41. Email / Meeting Intake Intelligence
42. Professional Client Delivery Workspace

Mine the older Agent Factory/Army/ZEUS artifacts for useful mechanisms including:

- semantic zoom
- direct manipulation
- macro human-swarm commands
- formation/topology interaction
- context logistics
- collision detection
- specialist deployment
- causal replay
- spatial memory
- uncertainty visualization

But treat those as **interaction research**, not permission to restore superseded branding or turn CELL OS into a military game.

9. CELL STUDIO

CELL OS actually builds artificial organizations.

Cell Studio must therefore become one of the product's signature experiences.

It needs to make these concepts understandable:

- formation/topology
- Operative seats
- deterministic workers
- model/runtime selection
- capabilities
- tools
- memory mounts
- context policies
- communication policy
- resource envelopes

- authority
- human gates
- evaluators
- evidence contract
- versioning
- certification

Compare:

1. freeform node graph
2. form-first editor
3. structured canvas + inspector
4. natural-language builder + generated configuration
5. hybrid

My current hypothesis is:

HYBRID

**natural language → generated Cell → structured canvas → inspector → machine-readable config
→ version diff → readiness → launch**

Do not choose a graph editor merely because graphs look agentic.

10. MOTION SYSTEM

Motion should communicate only:

- orientation
- causality
- status
- continuity
- feedback

Suggested timing system:

- press ~90ms
- hover ~120ms
- tooltip ~130ms
- popover ~160ms
- menu/tab ~180ms
- inspector ~240ms
- consequence sheet ~260ms
- workspace continuity ~320ms
- complex deliberate visualization <500ms

State transitions occur in place.

Do not make Working → Waiting → Done cause cards to jump around.

Pearl glisten may exist only in peripheral/empty chrome and should disappear during focused work.

Prefer **event-linked shimmer**—for example, a single restrained highlight when a mission becomes Verified—over constant decorative movement.

Explicitly design reduced-motion behavior.

11. HOVER AND OVERLAY GRAMMAR

Use:

Tooltip → short supplemental information

Popover → compact metadata or quick action

Inspector → deep state, relationships, provenance, context, versions, evidence and actions

Modal → bounded synchronous form

Consequence Sheet → consequential/high-risk action

Never hide a required instruction behind a tooltip.

Hover should reveal detail without causing layout shifts, card scaling or visual noise.

12. DATA VISUALIZATION

Do not design an analytics dashboard full of permanent charts.

CELL OS is an action system first.

Use compact state and metrics by default, then progressively disclose analysis.

Every important metric must have at least one:

- target
- baseline
- expected band
- comparison period
- annotation

Use:

- line → trend
- horizontal bar → rank

- bullet/target band → target
- timeline/lane → operational time
- table + sparklines → many Operatives/Cells
- graph → dependency/topology only
- Pareto scatter + linked table → Evolution

Every chart should lead back to the missions, runs and evidence producing the number.

13. MOBILE

Do not shrink desktop.

Mobile priorities:

1. Needs You
2. blocked / failed / risky
3. Since You Were Away
4. pinned mission/session status
5. evidence-aware decisions
6. voice briefing/navigation

Suggested bottom navigation:

Today · Inbox · Missions · More

Test:

- a session silently waiting for permission
 - one important decision among twenty working sessions
 - a deployment failure
 - returning after two hours
 - "Read me what needs me"
 - remotely pausing/approving a safe bounded action
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14. IA TOURNAMENT

Do not automatically accept the supplied IA.

Compare at least:

A — Switchboard/NERVE unified shell

B — Multi-app suite

C — Mission-centric object OS

D — Command-first adaptive workspace

Score each against:

- five-second comprehension
- learnability
- expert speed
- navigation burden
- discoverability
- spatial memory
- mobile
- Cell Studio integration
- implementation complexity
- ability to handle future capabilities

Select a winner and explain the decision.

15. DESIGN QUALITY GATE

Score every major design 1–10 for:

- five-second comprehension
- attention routing
- task speed
- interruption recovery
- trust calibration
- error prevention
- reversibility
- information architecture
- spatial memory
- accessibility
- premium aesthetic
- information density
- mobile supervisory value
- technical feasibility
- architectural truth

Automatically reject or redesign anything involving:

- focus theft
- hidden critical state
- chat as canonical product state
- color-only state
- critical hover-only information

- modal stacking
 - Done represented as Verified
 - dangerous side effect without preview
 - decorative motion required for understanding
 - an avatar/world mode maintaining separate state
 - a freeform node graph as the only critical configuration mechanism
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16. REQUIRED OUTPUT

Do not return a moodboard.

Return a buildable:

CELL OS PRODUCT DESIGN SYSTEM & EXPERIENCE SPECIFICATION

Containing:

1. Executive design thesis
2. Artifact reconciliation ledger
3. Current UX friction audit
4. Implementation-truth table
5. Product mental model
6. IA tournament
7. Selected IA
8. Desktop NERVE architecture
9. Mobile NERVE architecture
10. Page-by-page specifications
11. Cell Studio deep specification
12. Component library
13. Command/chat/voice grammar
14. Motion system
15. Microinteraction system
16. Hover/tooltip/inspector/modal grammar
17. Design tokens
18. Data-viz grammar
19. Empty/loading/error/degraded states
20. Accessibility and human-factors specification
21. Feature ranking: P0 / P1 / P2 / Research
22. Backend state/events required by each major feature
23. UX experiments and measurable success criteria
24. What NOT to build
25. Contradictions/unresolved decisions
26. Final high-fidelity product storyboard

The final storyboard must show:

Request arrives

- **Blackbox Intake**
- **context/scoping**
- **Mission Contract**
- **Cell compilation**
- **parallel safe execution**
- **Needs You**
- **evidence-aware human decision**
- **Verified Success**
- **artifact/client delivery**
- **Replay/Learning**
- **automation/distillation opportunity**

The finished product should feel like an:

iPhone-class operating experience for synthetic organizations

Simple on the surface.

Extraordinarily deep when inspected.

Fast enough for an expert.

Understandable enough for someone new to agentic systems.

Calm under high activity.

Beautiful without visual noise.

And trustworthy enough that the user understands not only **what the organization is doing**, but **why**, **under what authority**, **with what evidence**, and **what will happen next**.